

See discussions, stats, and author profiles for this publication at: <https://www.researchgate.net/publication/225741839>

Calculations of the percolation thresholds of a three-dimensional (icosahedral) Penrose tiling by the cubic approximant method

Article in Crystallography Reports · January 2005

DOI: 10.1134/1.2132400

CITATIONS

3

READS

842

2 authors, including:



Ruslan Mikhailovich Zakalyukin

Russian Academy of Sciences

78 PUBLICATIONS 643 CITATIONS

SEE PROFILE

Some of the authors of this publication are also working on these related projects:



SnF2@SWCNT nanocomposite [View project](#)



Fluoride glasses [View project](#)

LATTICE DYNAMICS AND PHASE TRANSITIONS

Calculations of the Percolation Thresholds of a Three-Dimensional (Icosahedral) Penrose Tiling by the Cubic Approximant Method

R. M. Zakalyukin and V. A. Chizhikov

*Shubnikov Institute of Crystallography, Russian Academy of Sciences,
Leninskiĭ pr. 59, Moscow, 119333 Russia*

e-mail: chizhikov@ns.crys.ras.ru

Received February 17, 2005

Abstract—The percolation thresholds of a three-dimensional Penrose tiling with icosahedral symmetry are determined using the cubic approximant method. The percolation thresholds of the three-dimensional Penrose tiling for the site problem and the bond problem are calculated with an accuracy of 0.001. © 2005 Pleiades Publishing, Inc.

INTRODUCTION

The percolation theory is one of the theoretical approaches that have long been in use for predicting a great variety of properties of materials [1, 2]. This theory has found wide application in many fields of research, specifically in the growth phenomena [3] and conduction processes [4], strength of materials, theory of phase transitions, etc. Applications of the percolation theory for solving problems in the physics of partially ordered and disordered media, as well as in the soft-matter physics, are of special interest.

The percolation theory and appropriate numerical methods have been developed continuously: available data on the percolation thresholds for even very simple structures, such as simple cubic, face-centered cubic, body-centered cubic, and other lattices, have been refined and complemented using calculations performed within the percolation theory under different conditions [5]. The percolation theory is an efficient tool for studying disordered structures [6]. In recent years, publications have appeared that are concerned with the percolation in structures of a new type, namely, quasi-periodic tilings. The first numerical experiments for two-dimensional decagonal Penrose tilings were carried out by Ziff and Babalievski [7]. However, there is an appreciable gap in the calculations of the percolation thresholds for three-dimensional quasi-periodic structures, which is associated primarily with the considerable difficulties encountered in performing computer experiments.

The purpose of this study was to calculate the percolation thresholds for the three-dimensional icosahedral Penrose tiling, which is of interest as a model object of investigation for several reasons. The three-dimensional Penrose tiling, like random disordered structures, is locally inhomogeneous but exhibits a long-

range quasi-periodic order. This tiling has an exact mathematical description. In particular, the coordination numbers of all sites and the mean coordination number can be determined exactly. Special interest expressed by researchers in the three-dimensional Penrose tiling is associated with the discovery of icosahedral quasicrystals whose structures have much in common with the tiling structure [8, 9].

ICOSAHERAL SYMMETRY AND ICOSAHERAL LOCAL ORDER

The point symmetry inherent in periodic crystals imposes specific requirements on the internal structure of the crystal, specifically on the local atomic order. In this case, the local order is considered to be understood as the nearest environment of each atom within one or several coordination spheres. The natural constraint placed on the structure is that, for any local atomic configuration and an arbitrary element of the symmetry point group, it is possible to find another atomic configuration related to the initial configuration by the given symmetry element. Consequently, a set of possible bonds between atoms in the structure can be described by a finite set of vectors whose point symmetry is no lower than the point symmetry of the crystal itself.

Unlike periodic crystals, in quasicrystals, as well as in amorphous materials, each atom has a unique nearest environment. Therefore, a set of possible bonds between the nearest atoms is an infinite set of vectors. This set of vectors in amorphous materials possesses spherical symmetry, whereas the symmetry of the set of interatomic distances in quasicrystals coincides with the point symmetry of the quasicrystal itself. It should be noted that the density of the set of interatomic distances reaches maxima at specific points that can be associated with particular ideal positions of the bonds.

This allows one to construct an idealized model of the quasicrystal in which the set of possible interatomic bonds, as in the crystal, is reduced to a finite set of vectors. It is worth noting that the smaller the number of vectors in the set, the more effective the model. It is obvious that the reduced set of interatomic distances should retain the symmetry of the quasicrystal.

Let us consider an idealized structure of an icosahedral quasicrystal. A set of bonds between the nearest neighbors forms a set of vectors that has icosahedral symmetry. An arbitrary vector related by the elements of the icosahedral symmetry group gives a set consisting of 120 vectors. This set seems too cumbersome. It is much more convenient to use specific vectors aligned parallel to the symmetry axes of the icosahedron. Earlier, it was established that the experimental data obtained for structures of icosahedral quasicrystals are in excellent agreement with the local order at which the nearest neighbor atoms are spaced at r_3 , r_5 , and r_2 intervals along the twofold, fivefold, and twofold axes, respectively [10, 11]. The ratios between these distances are as follows: $r_3 : r_5 : r_2 = \sqrt{3} : \sqrt{\tau + 2} : 2$, where

$\tau = \frac{\sqrt{5} + 1}{2} \approx 1.618$ is the golden mean. In the particular

case, the bonds r_3 are absent. Hence, the local order is referred to as the icosahedral order. Generally speaking, the icosahedral order can be described by introducing only the distances r_5 . In this case, the distances r_2 appear automatically as sums of the vectors r_5 . Examples of the icosahedral local order are provided by structures of an Al_{12}Mn crystal and three-dimensional Penrose tiling [12].

Expressions for the vectors r_5 have the simplest form in the coordinate system in which the Cartesian axes coincide with three mutually perpendicular twofold axes of the icosahedron. These 12 vectors, which are directed from the center to the vertices of the icosahedron, can be written in the form $(\pm 1, \pm\tau, 0)$ to within a scale factor. This formula accounts for the cyclic permutations of the coordinates, and the plus and minus signs can be chosen for each of the coordinates independently. In essence, the icosahedral local order is as follows: by going from neighbor to neighbor and using only the vectors r_5 , any two atoms of the structure can be connected by a chain of steps. If an arbitrary atom is placed at the origin of the coordinates, the coordinates of any other atom can be represented as a linear combination of the vectors r_5 with integral coefficients. However, proper allowance must be made for the fact that the vectors directed to opposite vertices of the icosahedron differ only in sign and, hence, the number of independent vectors is reduced to six (Fig. 1). Therefore, the coordinates of an arbitrary atom in the structure with

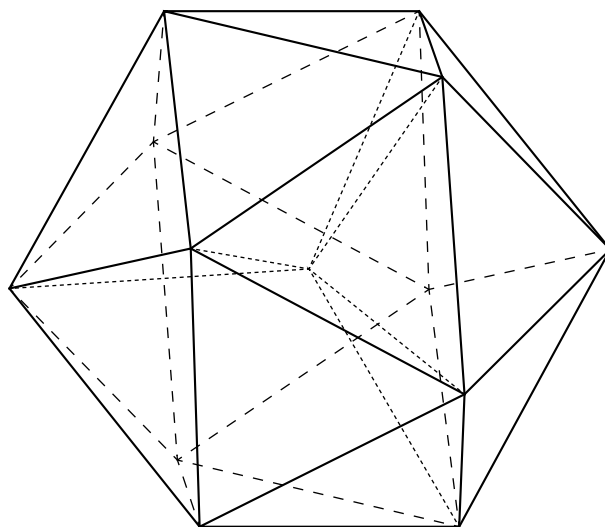


Fig. 1. Six basis vectors (dotted lines) of the structure with an icosahedral local order connect the icosahedron center with its vertices.

icosahedral local order can be written in the form

$$\mathbf{r} = n_1 \begin{pmatrix} 1 \\ \tau \\ 0 \end{pmatrix} + n_2 \begin{pmatrix} 0 \\ 1 \\ \tau \end{pmatrix} + n_3 \begin{pmatrix} \tau \\ 0 \\ 1 \end{pmatrix} + n_4 \begin{pmatrix} 1 \\ -\tau \\ 0 \end{pmatrix} + n_5 \begin{pmatrix} 0 \\ 1 \\ -\tau \end{pmatrix} + n_6 \begin{pmatrix} -\tau \\ 0 \\ 1 \end{pmatrix}. \quad (1)$$

By changing over to the matrix form, relationship (1) can be rewritten as

$$\mathbf{r} = E_{ph} \mathbf{n}, \quad E_{ph} = \begin{pmatrix} 1 & 0 & \tau & 1 & 0 & -\tau \\ \tau & 1 & 0 & -\tau & 1 & 0 \\ 0 & \tau & 1 & 0 & -\tau & 1 \end{pmatrix}, \quad (2)$$

where E_{ph} is the matrix specifying the projection of sites $\mathbf{n}$ of the six-dimensional hypercubic lattice onto the three-dimensional physical space and $\mathbf{n}$ and $\mathbf{r}$ are column vectors. In such a manner, we introduced the multidimensional formalism typical of the theory of quasicrystals [12–14].

In the framework of the multidimensional formalism, the physical space, i.e., the space in which the structure and its physical properties are determined, is treated as a subspace of the space of higher dimension (in our case, it is the six-dimensional space). The three-dimensional subspace complementary to the physical

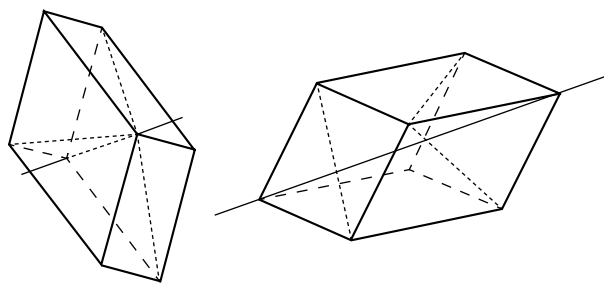


Fig. 2. Oblate (at the left) and prolate (at the right) Ammann rhombohedra. Thin solid lines indicate the threefold symmetry axes. The rhombohedron edges are the vectors r_5 . Dotted lines represent the short diagonals r_2 of the visible faces and the short internal diagonal r_3/τ of the oblate rhombohedron.

subspace is defined by the projection

$$\mathbf{r}_\perp = E_\perp \mathbf{n}, \quad E_\perp = \begin{pmatrix} \tau & 0 & -1 & \tau & 0 & 1 \\ -1 & \tau & 0 & 1 & \tau & 0 \\ 0 & -1 & \tau & 0 & 1 & \tau \end{pmatrix}. \quad (3)$$

This complementary subspace is called the perpendicular space of the icosahedral quasicrystal.

All the sites of the six-dimensional hypercubic lattice projected by the matrix E_{ph} densely occupy the physical space, much as all the sites projected by the matrix $E_\perp$ densely occupy the perpendicular space. In order to simulate structures with physically realistic atomic densities, it is necessary to decrease the number of sites of the six-dimensional lattice projected onto the physical space. A characteristic feature of icosahedral quasicrystals is that their projection onto the perpendicular space is a set with limited linear sizes. It is common practice to consider a bounded domain (or several disconnected domains) in the perpendicular space such that only those sites of the six-dimensional lattice whose projections $\mathbf{r}_\perp$ lie in this bounded domain belong to the structure. This domain is referred to as the acceptance domain. Therefore, the acceptance domain in the perpendicular space completely determines the atomic structure in the physical space. It should be noted that the acceptance domain of the structure with icosahedral symmetry should also have icosahedral symmetry.

THE THREE-DIMENSIONAL PENROSE TILING

A three-dimensional Penrose tiling has icosahedral symmetry. By analogy with the two-dimensional Penrose tiling composed of rhombi of two types (with acute angles of $\frac{2\pi}{5}$ and $\frac{2\pi}{10}$) [15, 16], the three-dimensional Penrose tiling consists of elementary polyhedra of two types, namely, the so-called Ammann rhombohedra [12]. The oblate and prolate Ammann rhombohedra are

parallelepipeds whose faces each are represented by identical rhombi with the acute angle $\alpha = \arccos \frac{1}{\sqrt{5}} \approx$

63.4° (Fig. 2). Each of the Ammann rhombohedra can be represented as a cube that is either oblate or prolate along one of the internal diagonals (with retention of the edge size), which explains their names. The volume of the prolate rhombohedron is larger than that of the oblate rhombohedron by a factor of τ .

Inside the tiling, the edges of the rhombohedra are aligned parallel to the fivefold axes of the icosahedron and, hence, can be related to the vectors r_5 . For example, the parallelepiped constructed by the vectors $(\tau, 0, 1)$, $(1, \tau, 0)$, and $(0, 1, \tau)$ is a prolate Ammann rhombohedron, whereas the parallelepiped constructed by the vectors $(\tau, 0, -1)$, $(-1, \tau, 0)$, and $(0, -1, \tau)$ is an oblate Ammann rhombohedron. Therefore, the three-dimensional Penrose tiling can be treated as a structure with icosahedral local order in which atoms are located at vertices of the rhombohedra.

The three-dimensional Penrose tiling is conveniently described in terms of the aforementioned multi-dimensional formalism. In this case, the acceptance domain has the form of a rhombic triacontahedron, which, in turn, can be represented by the projection of the six-dimensional hypercubic unit cell onto the perpendicular space (Fig. 3). The rhombic triacontahedron possesses the required icosahedral symmetry. This polyhedron contains 30 rhombic faces perpendicular to the twofold axes of the icosahedron. If the center of the triacontahedron is placed at the origin of the coordinates, its 32 vertices are specified by vectors of the type $(\pm\tau^2, \pm\tau, 0)$, $(\pm\tau, \pm\tau, \pm\tau)$, and $(\pm\tau^2, 0, \pm 1)$. These relationships take into account the cyclic permutations of the coordinates, and the plus and minus signs can be chosen for each coordinate independently.

Special attention should be focused on the boundary of the acceptance domain. When the projection of a particular site of the six-dimensional lattice onto the perpendicular space lies on one of the faces of the triacontahedron, the projection of another site should also be located at the corresponding position on the opposite face. For example, we consider two opposite rhombic faces of the triacontahedron in the planes $z_\perp = -\tau^2$ and $z_\perp = \tau^2$. (It is assumed that the center of the triacontahedron coincides with the origin of the coordinates.) In this case, the distance between the faces is determined by the vector $(0, 0, 2\tau^2)$ in the perpendicular space to which the vector $\Delta\mathbf{n} = (0, -1, 1, 0, 1, 1)$ corresponds in the six-dimensional space. Consequently, if the projection of the site $\mathbf{n}_b$ lies on the lower rhombus, the site $\mathbf{n}_t = \mathbf{n}_b + \Delta\mathbf{n}$ is projected onto the upper rhombus. The projections of these sites onto the physical space are separated by the distance $\left(0, 0, -\frac{2}{\tau}\right)$, which is not typical of three-dimensional Penrose tilings.

In order to exclude the appearance of surplus sites in the tiling, it is necessary to take into account the portion of the boundary belonging to the acceptance domain. However, there exists another method frequently used for eliminating frustrations: the triacontahedron as a whole can be shifted in such a way that its faces will not contain projections of the sites. In actual fact, although the projections of the sites of the six-dimensional lattice completely occupy the perpendicular space, their coordinates have the special form $i\tau + j$, where i and j are integral numbers. If the center of the acceptance domain is displaced from the origin of the coordinates, for example, to the point $(\pi, e, \sqrt{2})$, it is obvious that none of the sites will be projected onto the triacontahedron boundary.

In the general case, different positions of the triacontahedron center correspond to different tilings. In this respect, it is customary to speak about Penrose tilings in plural. However, without a loss in generality, we can speak about the Penrose tiling as a single structure if we consider that all the possible tilings are locally isomorphic to each other. The meaning of the term "local isomorphism" can be briefly explained as follows [17]. Two infinite structures are said to be locally isomorphic if any finite fragment of one structure can be found in another structure. When we are dealing with Penrose tilings, any fragment containing in one tiling is repeated an infinite number of times in each tiling and the distance from an arbitrary point of the tiling to the nearest repetition of the given fragment does not exceed its several linear sizes.

The local isomorphism plays an important role in the definition of the concept of point symmetry of quasi-periodic structures. Indeed, the lack of translational symmetry, which is characteristic of these structures, can be explained by the fact that their point groups involve elements, in particular, fivefold axes, which are incompatible with the periodicity. Such axes can be encountered in the structure only once for each of the possible directions. For example, the structure with icosahedral point symmetry can be characterized by no more than six fivefold axes aligned parallel to the six different directions. Furthermore, symmetry elements that transform the quasi-periodic structure into itself can be absent altogether. The question arises as to which argument counts in favor of the hypothesis that this structure has a symmetry point group. For mathematical rigor, the point symmetry element can be defined as follows: a structure has a point symmetry element if the related structure is locally isomorphic to the initial structure. It is in this sense that any three-dimensional Penrose tiling has icosahedral point symmetry.

SITES AND BONDS

Investigation into the structural properties within the percolation theory requires knowledge of the topology

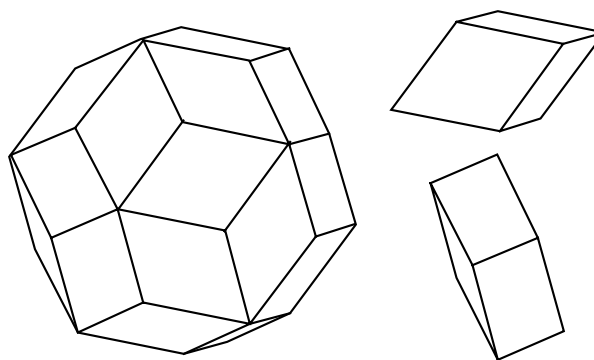


Fig. 3. Acceptance domain in the form of a rhombic triacontahedron for the three-dimensional Penrose tiling in the perpendicular space (at the left). The prolate (at the upper right) and oblate (at the lower right) Ammann rhombohedra are the respective projections of the oblate and prolate Ammann rhombohedra in the physical space onto the perpendicular space.

of the bonds between sites (atoms). In this case, other structural features, such as precise coordinates of sites, symmetry, or curvature of the enclosing space, are immaterial. For the three-dimensional Penrose tiling under consideration, the vertices of Ammann rhombohedra are naturally chosen as sites. Bonds between the sites can be chosen using different variants; however, in any case, the set of all bonds should include the set of all edges of the rhombohedra.

In the first variant, the bonds can be reduced to the edges of the Ammann rhombohedra. Then, the coordination numbers of specific sites vary from 4 to 12 (except for 11). The mean coordination number is 6. This holds true for any structures constructed from parallelepipeds when the edges are chosen as bonds. Actually, for one parallelepiped in the structure, there are one vertex and three edges. (This becomes evident if vertices and edges of the parallelepiped are represented in a "volume" form, i.e., vertices in the form of spheres and edges in the form of cylinders. The sum of internal solid angles at vertices of any parallelepiped is equal to 4π , which corresponds to one full sphere. At the same time, the sum of internal angles between the adjacent faces of the parallelepiped is equal to 6π , which corresponds to three circles in the bases of the cylinders.) Consequently, for one site in the structure, there are three bonds. Taking into account that each bond connects two sites, the desired mean coordination number can be determined as $\langle z \rangle = 6$.

In the initially chosen length units [see relationships (1), (2)], the edges of the Ammann rhombohedra have the length $r_5 = \sqrt{\tau + 2} \approx 1.902$. However, in the three-dimensional Penrose tiling, there are considerably shorter distances between a number of sites. These are vertices located at opposite ends of the short diagonal in the oblate Ammann rhombohedron (Fig. 2). The short internal diagonals of the oblate rhombohedra are parallel to the threefold axes and have the length $r_3/\tau =$

$\frac{\sqrt{3}}{\tau} \approx 1.070$; i.e., they are equal to almost half the edges. Therefore, it is reasonable to include these diagonals in the set of bonds and to consider such a set of sites and bonds as the second variant. In this case, the coordination numbers of specific sites vary from 6 to 12 (except for 11). In order to determine the change in the coordination number, it is necessary to calculate the ratio between the oblate and prolate Ammann rhombohedra.

There exists a general method for determining the frequency of occurrence of a particular fragment in a structure. First, it is necessary to construct the projection of this fragment onto the perpendicular space. In order for this fragment to be involved in the structure, the projection of the fragment should be located in the acceptance domain. The set of all possible displacements in the perpendicular space that form the projection of the fragment in the acceptance domain determines a specific region of finite volume V_{fr} . It is clear that the frequency of occurrence of the given fragment in the structure is proportional to the volume V_{fr} . Consequently, the problem of determining the frequency of occurrence of a particular fragment is reduced to a purely geometric problem.

The projection of the oblate Ammann rhombohedron onto the perpendicular space is a prolate Ammann rhombohedron and vice versa (Fig. 3). By displacing the projections inside the triacontahedron, it is possible to determine the volumes corresponding to the oblate ($V_{fr,o}$) and prolate ($V_{fr,p}$) rhombohedra. The ratio of these volumes is equal to the ratio between the numbers of prolate and oblate rhombohedra in the three-dimensional Penrose tiling $v_p/v_o = V_{fr,p}/V_{fr,o}$. This ratio is given by

$$v_p/v_o = \tau. \quad (4)$$

Therefore, the frequency of occurrence of prolate rhombohedra is higher than that of oblate rhombohedra by a factor of τ . Hence, it follows that the fraction of oblate rhombohedra is $v_o/(v_o + v_p) = \frac{1}{\tau^2}$. Note that each

of the oblate rhombohedra has only one short diagonal r_3/τ connecting two vertices. As a result, the change in the coordination number of sites is equal to $\frac{2}{\tau^2}$ and,

therefore, we have $\langle z \rangle = \left(6 + \frac{2}{\tau^2}\right) = (10 - 2\tau) \approx 6.764$.

The set of bonds can be complemented by short diagonals of the rhombic faces of the Ammann rhombohedra (the third variant). These diagonals are aligned parallel to the twofold axes and have the length $r_2 = 2$ (Fig. 2); i.e., they are only slightly longer than the edges: $r_2/r_3 \approx 1.051$. For one rhombohedron in the structure, there are three rhombic faces and, hence,

three diagonals r_2 . Consequently, the mean coordination number increases by 6: $\langle z \rangle = (16 - 2\tau) \approx 12.764$. Note that the coordination numbers of specific sites take on values from 11 to 19.

Two oblate Ammann rhombohedra can share a face. In this case, they share a vertex from which two short internal diagonals r_3/τ of these rhombohedra emerge. The distance between the other two ends of these diagonals is also equal to r_2 . The inclusion of these distances in the set of bonds is not justified physically because one more site is located at the midpoint between two sites almost in the same line with them: the angle between the short diagonals of the adjacent oblate Ammann rhombohedra is equal to

$$\arccos\left(-\frac{\sqrt{5}}{3}\right) \approx 138.2^\circ. \text{ In general, among the three}$$

aforementioned sets of bonds, the first variant is of formal mathematical interest, whereas the last variant is most justified physically.

CUBIC APPROXIMANTS

The irrational number τ can be approximated by rational relationships. In this situation, the best approximation can be achieved with natural numbers of the Fibonacci series. This number sequence, which was originally derived as the solution of the rabbit reproduction problem, is determined by the recurrent relation

$$f_{n+1} = f_n + f_{n-1}, \quad (5)$$

where $f_1 = f_2 = 1$. The first terms of this series of natural numbers are as follows:

$$1, 1, 2, 3, 5, 8, 13, 21, 34, 55, \dots$$

The n th Fibonacci number is given by the formula

$$f_n = \frac{1}{\sqrt{5}} \left[\tau^n - \left(-\frac{1}{\tau}\right)^n \right]. \quad (6)$$

This formula immediately leads to the limiting relationship

$$\lim_{n \rightarrow \infty} \frac{f_n}{f_{n-1}} = \tau. \quad (7)$$

The relationship between the Fibonacci numbers and the golden mean

$$\tau^n = f_n \tau + f_{n-1} \quad (8)$$

turned out to be very useful and has often been used to simplify mathematical calculations. Note that, if the Fibonacci series of natural numbers is complemented by the terms $f_0 = 0$ and $f_{-n} = (-1)^{n+1} f_n$, relationship (8) will be valid for all integral exponents of the number τ . It is interesting to note that the new terms of the extended Fibonacci series also satisfy the recurrent relation (5) and expression (6) for the n th term.

Another important property of the Fibonacci numbers is associated with their parity. The greatest common divisor of two sequential Fibonacci numbers is equal to unity. This can be easily demonstrated by *reductio ad absurdum*. Indeed, let us assume that two sequential terms of the series f_n and f_{n-1} have a common divisor $k \geq 2$. Hence, it follows from the recurrent relation (5) that any Fibonacci number can be divided by k , which is obviously not true. In particular, the given property implies that two sequential Fibonacci numbers cannot be even simultaneously. It can be readily seen that even (e) and odd (o) Fibonacci numbers alternate with a period of 3 (oee). As will be shown below, the parity of the Fibonacci numbers affects the symmetry of cubic approximants.

The noncrystallographic point symmetry of the quasicrystals can be achieved by specially choosing the perpendicular space $E_\perp$. By changing the space $E_\perp$, one can obtain structures with other symmetries, including periodic structures. The main constraint, as before, is that the projection of the structure onto the perpendicular space should have limited linear sizes. Note that the physical space remains the same. This ensures retention of the initial icosahedral local order.

The situation in which a new perpendicular space is obtained from projection (3) through a simple replacement of the number τ by the rational fraction f_n/f_{n-1} is of particular interest. After this replacement, the projection of the sites of the six-dimensional hypercubic lattice onto the new perpendicular space can be defined by the matrix

$$E_\perp^n = \frac{1}{f_{n-1}} \begin{pmatrix} f_n & 0 & -f_{n-1} & f_n & 0 & f_{n-1} \\ -f_{n-1} & f_n & 0 & f_{n-1} & f_n & 0 \\ 0 & -f_{n-1} & f_n & 0 & f_{n-1} & f_n \end{pmatrix}. \quad (9)$$

The three-dimensional space $E_\perp^n$ determines a cubic periodic structure, the so-called Fibonacci crystal of the n th order. The Fibonacci crystals are often referred to as cubic approximants of icosahedral quasicrystals. Many cubic approximants of low orders have been revealed in alloys similar in composition to quasicrystals. In the largest known approximants, the number of atoms in a unit cell can be as large as several hundreds. The basic symmetry properties of the Fibonacci crystals [18, 19] are given below without proof.

Projection of a Six-Dimensional Lattice onto the Space $E_\perp^n$

In the case of icosahedral symmetry, the projections of the sites of the six-dimensional hypercubic lattice completely occupy the space $E_\perp$. A different situation occurs with Fibonacci crystals. Actually, it follows from matrix (9) that the sites of the six-dimensional lat-

tice are projected onto the space $E_\perp^n$ into sites of a simple cubic lattice with spacing $\frac{1}{f_{n-1}}$. However, this does

not necessarily mean that all sites of the simple cubic three-dimensional lattice are the projections of the sites of the six-dimensional lattice. In this situation, an important role is played by the parity of the Fibonacci numbers f_n and f_{n-1} corresponding to the n th order of the approximant. If one of the numbers f_n and f_{n-1} is an even number (the cases of *e/o* and *o/e*), the projection of the six-dimensional hypercubic lattice onto the space $E_\perp^n$ occupies all the sites of the simple cubic lattice with spacing $\frac{1}{f_{n-1}}$. However, when both f_n and f_{n-1} are odd numbers (the case of *o/o*), the projection of the six-dimensional hypercubic lattice onto the space $E_\perp^n$ occupies one face-centered cubic sublattice of the simple cubic lattice or, to put it differently, form a face-centered cubic lattice with spacing $\frac{2}{f_{n-1}}$.

Periodicity

The projection of the six-dimensional hypercubic lattice onto the space $E_\perp^n$ is degenerate: to each point of the projection there corresponds an infinite set of the sites of the six-dimensional lattice. The criterion that a particular site belongs to the structure is the location of its projection in the acceptance domain. Therefore, all the sites projected into the same point either belong to the structure or do not belong to it (in principle, it is possible to analyze the exceptions from the formulated rule: this leads to the formation of Fibonacci crystals with double, triple, and similar unit cells). In order for two sites $\mathbf{n}_1$ and $\mathbf{n}_2$ of the six-dimensional lattice to be projected into the same point of the space $E_\perp^n$, they should obey the obvious equation

$$E_\perp^n \Delta \mathbf{n} = 0. \quad (10)$$

Here, we introduced the designation $\Delta \mathbf{n} = (\mathbf{n}_2 - \mathbf{n}_1)$. In the cases of *e/o* and *o/e*, the solution of Eq. (10) has the form

$$\Delta \mathbf{n} = m_x \mathbf{n}_x + m_y \mathbf{n}_y + m_z \mathbf{n}_z. \quad (11)$$

Here, m_x , m_y , and m_z are integral numbers and $\mathbf{n}_x$, $\mathbf{n}_y$, and $\mathbf{n}_z$ are six-dimensional vectors with the following integral coordinates:

$$\begin{aligned} \mathbf{n}_x &= (f_{n-1}, 0, f_n, f_{n-1}, 0, -f_n), \\ \mathbf{n}_y &= (f_n, f_{n-1}, 0, -f_n, f_{n-1}, 0), \\ \mathbf{n}_z &= (0, f_n, f_{n-1}, 0, -f_n, f_{n-1}). \end{aligned} \quad (12)$$

Therefore, the vectors $\mathbf{n}_x$, $\mathbf{n}_y$, and $\mathbf{n}_z$ can be treated as spacings of the Fibonacci crystals in the six-dimensional space and their projections onto the physical space $\mathbf{p}_i = E_{ph}\mathbf{n}_i$, where $i \in \{x, y, z\}$, can be treated as spacings of the Fibonacci crystals in the physical space. From relationships (2) and (12) with the use of equality (8), it is easy to find that $\mathbf{p}_x = (2\tau^n, 0, 0)$, $\mathbf{p}_y = (0, 2\tau^n, 0)$, and $\mathbf{p}_z = (0, 0, 2\tau^n)$; i.e., the Fibonacci crystals in the cases *e/o* and *o/e* have a primitive cubic structure with spacing $2\tau^n$. In the case of *o/o*, there arise an additional spacing $\frac{1}{2}(\mathbf{n}_x + \mathbf{n}_y + \mathbf{n}_z)$ in the six-dimensional space and the corresponding spacing (τ^n, τ^n, τ^n) in the physical space; i.e., the Fibonacci crystals have a body-centered cubic structure with the cubic unit cell parameter $2\tau^n$.

Acceptance Domain

In contrast to the case of icosahedral quasicrystals for which the acceptance domain is completely occupied by projection points, the Fibonacci crystals are characterized by the acceptance domain with a finite number of projection points; more precisely, the number of points in the acceptance domain is equal to the number of atoms in the unit cell of the Fibonacci crystal. In principle, this circumstance makes it possible to do without an acceptance domain: in order to determine the structure completely, it is sufficient to specify a particular number of points on a cubic (simple or face-centered) lattice in the space $E_{\perp}^n$. Nonetheless, the acceptance-domain formalism remains useful for a number of reasons. First, high-order approximants can contain a very large number of atoms in the unit cell whose simple enumeration is always extremely tedious and useful only rarely. Second, the Fibonacci crystals can bear a closer structural resemblance to the icosahedral crystal only in the case where their projection onto the perpendicular space $E_{\perp}^n$ will also resemble the projection of the quasicrystal onto the space $E_{\perp}$. In this respect, it is desirable to adapt the acceptance domain of the ideal quasicrystal to the Fibonacci crystals. Actually, for cubic approximants of the three-dimensional Penrose tiling, the role of the acceptance domain is played by a distorted rhombic triacontahedron whose 32 vertices are specified by the vectors $\left(\pm\frac{f_{n+1}}{f_{n-1}}, \pm\frac{f_n}{f_{n-1}}, 0\right)$, $\left(\pm\frac{f_n}{f_{n-1}}, \pm\frac{f_n}{f_{n-1}}, \pm\frac{f_n}{f_{n-1}}\right)$, and $\left(\pm\frac{f_{n+1}}{f_{n-1}}, 0, \pm 1\right)$ drawn from the center of the polyhedron. Unlike the coordinates of the vertices of the ideal rhombic triacontahedron, these coordinates are obtained with the replacements $\tau \rightarrow \frac{f_n}{f_{n-1}}$ and $\tau^2 \rightarrow \frac{f_{n+1}}{f_{n-1}}$.

Point Symmetry

The cubic approximants of icosahedral quasicrystals belong to the tetrahedral class *T* rather than to the cubic class *O*. In actual fact, unlike the cube, the icosahedron has no fourfold axes. As a result, the tetrahedral group rather than the cubic group is the icosahedral symmetry subgroup. In order for the Fibonacci crystal to possess tetrahedral symmetry, the projection of the crystal onto the perpendicular space $E_{\perp}^n$ should also have tetrahedral symmetry. Since this projection consists of a finite number of points that belong to the simple cubic or face-centered cubic lattice in the perpendicular space $E_{\perp}^n$, it is clear that the center of the projection and, hence, the center of the acceptance domain can be located only at points of a special type. These points for the simple cubic lattice (cases *e/o*, *o/e*) are represented by the origin of the cell $(0, 0, 0)$ and the center of the cell $\left(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right)$, whereas the corresponding points for the face-centered lattices (case *o/o*) have the coordinates $(0, 0, 0)$, $\left(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right)$, and $\left(\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right)$ (here, for convenience, the spacings of the cubic lattices are taken to be equal to unity, as is customary in crystallography). All the other special points are equivalent to the above points; i.e., they are related to them by the symmetry operations of the cubic lattices.

Spatial Symmetry

The symmetry space group of the *n*th-order Fibonacci crystal depends on the parity of the Fibonacci numbers f_n and f_{n-1} and also on the location of the center of the crystal projection in the perpendicular space. All the possible space groups of the Fibonacci crystals were determined earlier. The groups with the highest symmetry for all the cases under consideration are as follows: $Pm\bar{3}$ [cases *e/o*, *o/e*; the center of the projection is located at the point $(0, 0, 0)$], $Pa\bar{3}$ [cases *e/o*, *o/e*; the center of the projection is located at the point $\left(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right)$], $Im\bar{3}$ [case *o/o*, the center of the projection is located at the point $(0, 0, 0)$ or the point $\left(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right)$], and $I2_13$ [case *o/o*, the center of the projection is located at the point $\left(\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right)$]. The other possible groups with lower symmetry for the Fibonacci crystals are subgroups of the aforementioned groups.

Approximation of the Quasicrystal Structure by Approximants

In the case where the order n of the cubic approximant increases by unity, the unit cell parameter increases by a factor of τ and the unit cell volume increases by a factor of $\tau^3 \approx 4.236$. Hence, the number of atoms in the unit cell increases by a factor of more 4. Note that, on a scale of the order of one unit cell, the structure of the approximant coincides with the structure of the ideal quasicrystal. As a result, the quasicrystal can be considered the limit of the sequence of cubic approximants at $n \rightarrow \infty$. The approximants, as if, approach the quasicrystal (approximate the quasicrystal) in the structure and physical properties, which explains their names. For example, the diffraction pattern of the approximants with an increase in the order n becomes similar in relative intensities and positions of the Bragg reflections to the diffraction pattern of the quasicrystal. Beginning with a specific value of n , which is associated with the experimental error, the approximants become virtually identical to the quasicrystals [20].

The structural relation to icosahedral quasicrystals, on the one hand, and the periodicity, on the other hand, make the Fibonacci crystals convenient model objects for use in investigating the properties of the quasicrystals themselves. Specifically, the local atomic structure of quasicrystals can be more easily understood and described when the low-order cubic approximants whose atomic structures are determined exactly are known. Moreover, many mathematical statements regarding the ideal quasicrystals can be more readily proved first for approximants and then can be generalized with the use of the limit $n \rightarrow \infty$.

For example, let us prove relationship (4) for the three-dimensional Penrose tiling by using the Fibonacci crystals. The unit cell volume for the n th-order Fibonacci crystal can be written as $V_{cell} = (2\tau^n)^3 = 8\tau^{3n}$. We assume that the Fibonacci crystal is composed of structural units identical to the structural units of the Penrose tiling, i.e., oblate and prolate Ammann rhombohedra. The volume of the prolate Ammann rhombohedron constructed using the vectors $(\tau, 0, 1)$, $(1, \tau, 0)$, and $(0, 1, \tau)$ and the volume of the oblate Ammann rhombohedron constructed using the vectors $(\tau, 0, -1)$, $(-1, \tau, 0)$, and $(0, -1, \tau)$ are as follows:

$$V_p = \begin{vmatrix} \tau & 0 & 1 \\ 1 & \tau & 0 \\ 0 & 1 & \tau \end{vmatrix} = \tau^3 + 1 = 2\tau + 2 = 2\tau^2$$

and

$$V_o = \begin{vmatrix} \tau & 0 & -1 \\ -1 & \tau & 0 \\ 0 & -1 & \tau \end{vmatrix} = \tau^3 - 1 = 2\tau.$$

The volume ratio is equal to the irrational number $V_p/V_o = \tau$. As a result of this incommensurability, the unit cell volume of the Fibonacci crystal can be uniquely determined by the volumes of the oblate and prolate Ammann rhombohedra. Actually, we have

$$\begin{aligned} V_{cell} &= 8\tau^{3n} = 4f_{3n-1} \times 2\tau^2 + 4f_{3n-2} \times 2\tau \\ &= 4f_{3n-1}V_p + 4f_{3n-2}V_o. \end{aligned}$$

By changing over to the three-dimensional Penrose tiling at $n \rightarrow \infty$, we obtain relationship (4); that is,

$$v_p/v_o = \lim_{n \rightarrow \infty} \frac{4f_{3n-1}}{4f_{3n-2}} = \tau.$$

DETAILS OF THE COMPUTER EXPERIMENT

The percolation thresholds of the three-dimensional Penrose tiling were determined using computer simulation of the percolation in unit cells of the cubic approximants of the Penrose tiling. The Fibonacci crystals of the sixth, seventh, ninth, and tenth orders with space group $Pa\bar{3}$ were chosen as the objects of investigation. Table 1 presents the main characteristics of the unit cells in these crystals, such as the total number of sites in the cell, the number of sites on each face, and the numbers of prolate (n_p) and oblate (n_o) Ammann rhombohedra per unit cell. It can be seen from this table that the total number of rhombohedra ($n_p + n_o$) coincides with the number of sites. The ratio n_p/n_o is given for comparison with the ratio $v_p/v_o = \tau$ in order to illustrate the degree of similarity of the structure of the approximant to the ideal structure of the three-dimensional Penrose tiling.

In the percolation theory, there exist two different problems, namely, the site problem and the bond problem. In the site problem, each site either can be occupied (with probability p) or can be empty (with probability $1 - p$). If the site is empty, all bonds of this site are assumed to be dangling. In the bond problem, all sites are considered to be occupied, whereas the bonds either can connect the sites (with probability p) or can be dangling (with probability $1 - p$). A criterion for percolation in an infinite structure is the existence of an infinite cluster consisting of occupied sites that are connected to each other. Computer simulation experiments have been performed with bounded structural regions for which, historically, the natural criterion for percolation is the existence of a connected path between the opposite boundaries of the region under consideration.

In our case, we assumed that the cell of the cubic approximant percolates if there exists a connected path between any occupied site belonging to one of the faces (for example, the upper face) and any occupied site belonging to the opposite (lower) face. The periodic boundary conditions were imposed on the other (four) faces. The site of the cell was considered to belong to

Table 1. Characteristics of the unit cells of the cubic approximants of the three-dimensional Penrose tiling with space group $P\bar{a}3$

n	f_n/f_{n-1}	Number of sites		Number of Ammann rhombohedra		$(n_p/n_o) - \tau$
		in the cell	on the face	prolate (n_p)	oblate (n_o)	
6	8/5	10336	466	6388	3948	4.6×10^{-7}
7	13/8	43784	1220	27060	16724	-2.6×10^{-8}
9	34/21	785672	8362	485572	300100	-7.9×10^{-11}
10	55/34	3328160	21892	2056916	1271244	4.4×10^{-12}

the face when this site had at least one bond that emerged from the cell and intersected the given face.

Let $P(p)$ be the statistical probability that the structure percolates at a specified occupation number p . It is evident that the statistical probability satisfies the conditions $P(0) = 0$ and $P(1) = 1$. For a bounded structural region, $P(p)$ is a smooth function, which gradually increases when the occupation number p takes on values from 0 to 1. By contrast, for an infinite structure, the function $P(p)$ has a specific feature: the interval $[0, 1]$ can be separated into the two subintervals $[0, p_c)$ and $(p_c, 1]$ in such a way that $P(p \in [0, p_c)) = 0$ and $P(p \in (p_c, 1]) = 1$ and, at the point p_c , which is referred to as the percolation threshold, the value of $P(p)$ changes jumpwise from 0 to 1. For a number of simple two-dimensional structures, the percolation threshold can be determined by analytical methods. However, for the majority of structures, it is necessary to perform numerical experiments by the Monte Carlo method. The experimentally obtained functions $P(p)$ for bounded structural regions depend substantially on the region size. As the linear sizes of the cells under investigation increase, the graph $P(p)$ becomes more and more similar in appearance to a step, thus approaching the dependence $P(p)$ for an infinite structure. A comparison of the dependences $P(p)$ for cells of different sizes makes it possible to reveal the general regularities and to determine the percolation threshold with a required accuracy.

Now, we describe a direct algorithm for calculating the statistical probability P at a specified occupation number p for the site problem. Since any site either can be occupied or can be empty, for an arbitrary cell with N sites, there exist 2^N possible configurations with the probability $p^k(1-p)^{N-k}$, where k is the number of occupied sites. Then, the percolation probability can be calculated from the relationship

$$P(p) = \sum' p^k (1-p)^{N-k}, \quad (13)$$

where the prime on the sum sign denotes that the summation is performed only over the configurations for which the cell percolates.

Direct calculation of sum (13) cannot be accomplished in practice because of the very large number of configurations involved. In rare cases, it has been possible to apply some mathematical tricks to the calculation

of the percolation probability $P(p)$ without direct summation. However, for the vast majority of systems, it is necessary to use the Monte Carlo method and to determine the percolation probability $P(p)$ with a small random sample from the total number of configurations.

Let $p \in (0, 1)$ be a specified value of the occupation number. The configuration is constructed as follows: for each site of the cell, we determine in a random manner (using a random number generator) whether this site is occupied (with probability p) or is empty (with probability $1-p$). This configuration either percolates or does not percolate. Similarly, we construct M configurations, among which m configurations turn out to percolate. At $M \rightarrow \infty$, the ratio m/M tends toward the percolation probability; that is,

$$P(p) = \lim_{M \rightarrow \infty} \frac{m}{M}. \quad (14)$$

Moreover, it is advisable to know the root-mean-square deviation $\Delta P(p)$ of the percolation probability calculated according to relationship (14) from its true value determined from expression (13). The integral number m can take on values from 0 to M ; the probability that the integer m will take on a particular value, apparently, can be found from the formula $C_M^m P^m (1-P)^{M-m}$,

where $C_M^m = \frac{M!}{m!(M-m)!}$ is the number of combinations of M things taken m at a time. Therefore, the mean error of the computer experiment can be determined from the relationship

$$\Delta P(p) = \sqrt{\sum_{m=0}^M C_M^m P^m (1-P)^{M-m} \left(\frac{m}{M} - P \right)^2}$$

or, after simple mathematical transformations, from the expression

$$\Delta P(p) = \sqrt{\frac{P(p)(1-P(p))}{M}}. \quad (15)$$

ANALYSIS OF THE RESULTS OF THE COMPUTER SIMULATION AND THE PERCOLATION THRESHOLDS

The computer simulation of the percolation was carried out for six cases: the number of bonds sequentially included the edges of the Ammann rhombohedra (case I), the short internal diagonals of the oblate Ammann rhombohedra (case II), and the short diagonals of the faces of the Ammann rhombohedra (case III); moreover, the site problem and the bond problem were considered separately. In each case, the occupation number p took on specified values in successive steps $\Delta p = 0.001$ and the number of experiments performed at each point was $M = 1000$.

Figure 4 shows the characteristic dependences $P(p)$ obtained from the computer experiments. Four graphs in this figure correspond to Fibonacci crystals of the sixth, seventh, ninth, and tenth orders. It can be seen that, as the order n of the cubic approximant increases, the graph $P(p)$ becomes more and more similar in appearance to a step. It is also worth noting that points of pairwise intersections of all four graphs fall in a narrow range both along the abscissa axis and along the ordinate axis. Visually, it seems that all these graphs "meet" at a point in the vicinity of a particular value of the occupation number p . At $n \rightarrow \infty$, the percolation probability P at the given point remains unchanged, tends toward zero to the left of this point, and tends toward unity to the right of it. Now, it is not difficult to understand that this value of the occupation number is the percolation threshold p_c .

Of course, it is not to be expected that, with an increase in the accuracy of the experiment, all the graphs should actually intersect at the same point. Most likely, the pairwise intersections of the graphs will form a set of points $S(n, m)$ in the graph plane, where n and m are the orders of Fibonacci crystals ($n \neq m$). In this case, the percolation threshold will be determined as the abscissa of the limiting point of this set of points $S(n, m)$ with the quantities n and m tending toward infinity.

Nonetheless, even from these graphs, the percolation threshold can be determined with a certain accuracy dictated by the error of the experiment itself. Indeed, the points of intersections are arranged inside a rectangle whose base has a length close to the experimental step $\Delta p = 0.001$ and whose height is less than $6\Delta P$. Here, $\Delta P \approx 0.0158$ is the root-mean-square deviation of the percolation probability, which was calculated from relationship (15) for $P = 1/2$. (In this case, we can use the closeness of the discrete binominal distribution $C_M^m P^m (1 - P)^{M-m}$ to the continuous normal

distribution $\frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(m/M - P)^2}{2\sigma^2}\right)$ of the quantity (m/M) with a mean value P and a root-mean-square deviation $\sigma = \Delta P$. For a normal distribution, there exists

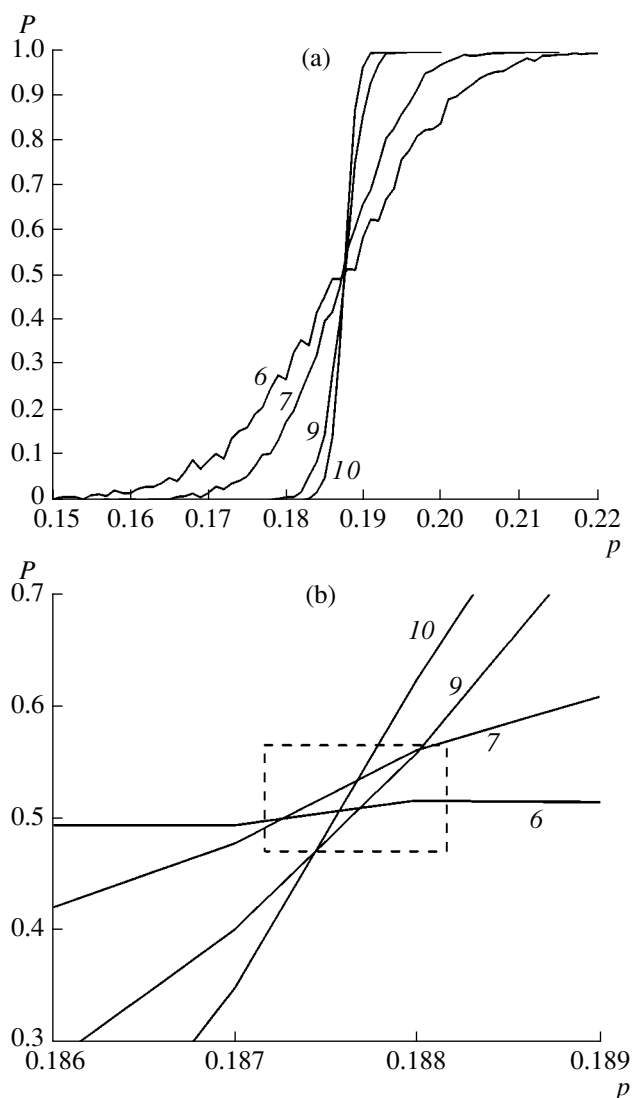


Fig. 4. Dependences of the percolation probability P on the occupation number p according to the computer simulation of the percolation for case III and the site problem (the graphs correspond to Fibonacci crystals of the sixth, seventh, ninth, and tenth orders): (a) the general view of the dependences and (b) the region of intersection of the graphs. The dashed rectangle of size $\Delta P \times 6\Delta P$ corresponds to the experimental error (it can be seen that the scatter of intersection points is of the order of the experimental error).

a nonnormal rule, the so-called three-sigma estimate, according to which the great bulk of the distribution is concentrated inside the domain $[P - 3\sigma, P + 3\sigma]$.) Thus, it is evident that the discrepancy between the points $S(n, m)$ does not exceed the experimental error and, hence, the percolation threshold can be considered to be determined accurate to within $\Delta p = 0.001$.

The percolation thresholds determined accurate to within $\Delta p = 0.001$ for the aforementioned six cases under investigation are presented in Table 2. These results are justified logically. It can be seen that an increase in the number of bonds in the structure leads to

Table 2. Percolation thresholds of the three-dimensional Penrose tiling

Included bonds	Mean coordination number	Site problem	Bond problem
Case I: Edges (r_5) of the Ammann rhombohedra	6	0.285	0.225
Case II: Edges of the Ammann rhombohedra and short internal diagonals (r_3/τ) of the oblate Ammann rhombohedra	$6 + \frac{2}{\tau^2} \approx 6.764$	0.271	0.207
Case III: Edges of the Ammann rhombohedra, short internal diagonals (r_3/τ) of the oblate Ammann rhombohedra, and short diagonals (r_2) of the faces of the Ammann rhombohedra	$12 + \frac{2}{\tau^2} \approx 12.764$	0.188	0.111

a decrease in the percolation threshold. On the other hand, the local inhomogeneity in the given case also brings about a decrease in the percolation threshold. In particular, for case I, as well as for a simple cubic lattice, the mean coordination number is 6. However, the percolation thresholds in the three-dimensional Penrose tiling appear to be substantially lower: 0.285 for the site problem (0.312 for the cubic lattice) and 0.225 for the bond problem (0.249 for the cubic lattice). Moreover, the following approximate relationship holds for the bond problem: $\langle z \rangle p_c \approx \text{const}$, where $\langle z \rangle p_c$ is equal to 1.35 for case I, 1.40 for case II and 1.42 for case III.

ACKNOWLEDGMENTS

The authors acknowledge the support of the Russian Science Foundation. This work was supported in part by the Russian Foundation for Basic Research (project nos. 02-02-16275-a and 05-02-16763-a), the Council for Grants of the President of the Russian Federation for Support of Leading Scientific Schools of the Russian Federation (grant no. NSh-1954.2003.2), and the Ministry of Education and Science of the Russian Federation within the program "Development of the Scientific Potential of the Higher School" (project no. III-3-4851).

REFERENCES

1. D. Stauffer and A. Aharony, *Introduction to Percolation Theory* (Taylor and Francis, London, 1992).
2. G. Grimmett, *Percolation and Disordered Systems* (Springer-Verlag, Berlin, 1997).
3. M. Kleman and O. B. Lavrentovich, *Soft Matter Physics: Introduction* (Springer-Verlag, Berlin, 2003).
4. B. I. Shklovskii and A. L. Efros, *Electronic Properties of Doped Semiconductors* (Nauka, Moscow, 1979; Springer-Verlag, New York, 1984).
5. S. P. Friedman and N. A. Seaton, *Transp. Porous Media* **30**, 241 (1998).
6. G. Schliecker and C. Kaiser, *Physica A* (Amsterdam) **269**, 189 (1999).
7. R. M. Ziff and F. Babalievski, *Physica A* (Amsterdam) **269**, 201 (1999).
8. D. Shechtman, I. Blech, D. Gratias, and J. W. Cahn, *Phys. Rev. Lett.* **53**, 1951 (1984).
9. D. Levine and P. J. Steinhardt, *Phys. Rev. Lett.* **53**, 2477 (1984).
10. V. E. Dmitrienko, *Acta Crystallogr., Sect. A: Found. Crystallogr.* **50**, 515 (1994).
11. V. E. Dmitrienko, *Mater. Sci. Forum* **150–151**, 199 (1994).
12. P. Kramer and R. Neri, *Acta Crystallogr., Sect. A: Found. Crystallogr.* **40**, 580 (1984).
13. P. A. Kalugin, A. Yu. Kitaev, and L. S. Levitov, *Pis'ma Zh. Éksp. Teor. Fiz.* **41**, 119 (1985) [*JETP Lett.* **41** (3), 145 (1985)].
14. M. Duneau and A. Katz, *Phys. Rev. Lett.* **54**, 2688 (1985).
15. R. Penrose, *Intelligencer* **2**, 32 (1979).
16. M. Gardner, *Penrose Tiles to Trapdoor Ciphers* (Freeman, New York, 1989; Mir, Moscow, 1993).
17. D. Levine and P. J. Steinhardt, *Phys. Rev. B* **34**, 596 (1986).
18. K. Niizeki, *J. Phys. A: Math. Gen.* **25**, 1843 (1992).
19. V. A. Chizhikov, *Kristallografiya* **44** (6), 814 (1999) [*Crystallogr. Rep.* **44** (6), 1024 (1999)].
20. V. E. Dmitrienko, *J. Phys. (France)* **51**, 2717 (1990).

Translated by O. Borovik-Romanova